# Theoretical Supplement

Information-theoretic foundations of an entropic systemic-stress indicator: what the main paper leaves out

Supplement to "Entropy Jumps and Information Relaxation in Financial Markets"

October 2026

**Abstract**

The main paper is refocused on a single object aimed at central banks: the dependence channel $-\frac{1}{2}\log\det R$ of a multi-market panel, its exact decomposition into within-block and between-block terms, and the pricing in nats of published stress scenarios. This document collects everything that refocusing removes from the body of the paper, to serve as an appendix or to be supplied on request. It contains the bridge from maximum entropy in volume time to geometric Brownian motion, the decomposition of entropy into three channels and the structural index $\mathcal{J} = D_{KL}(p \,\|\, \prod_i q_i)$, the jump-diffusion dynamics (one theorem, two postulates, and the Ornstein–Uhlenbeck structure forced by Doob's theorem), the empirical programme H1–H8 with its verdict against the calibrated nulls, and the full geometry of Kullback–Leibler-ball scenarios. Two results are new relative to arXiv v1. The first is a block decomposition identity for the dependence channel, which links this document to the main paper. The second shows that the full-dimensional lift of a portfolio scenario that minimises the ambient divergence costs exactly the radius $\eta$ and has an asset-by-asset reading (the shock is carried by each asset's covariance with the portfolio), while the default uniform dilation costs $N$ times as much in variance. The second point corrects a remark in v1.

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## Foreword: what this document contains, and why

**How it relates to the main paper.** The October 2026 assessment found that a single result of arXiv v1 clears both calibrated nulls on both panels: the regime gap of the dependence channel $H^{dep} = \frac{1}{2}\log\det R$. Phase A confirmed it in three forms (raw, Forbes–Rigobon devolatilised, rank-based) and showed that the absorption ratio of Kritzman et al. (2011), the benchmark measure of the same family, fails the block null on both panels. The main paper is therefore rebuilt around an "entropic CISS" in the sense of Holló, Kremer & Lo Duca (2012): the dependence channel of a multi-market panel, its within/between-block decomposition, central-bank-style validation, and the pricing of published scenarios in nats.

Everything else leaves the body of the paper. Not because it is wrong: the mathematical results remain exact. It is because these elements are not needed for the indicator, or because they do not survive the nulls and therefore cannot ground a surveillance tool. Collecting them here serves two purposes. A referee who asks where the dependence channel comes from, and why it is this channel rather than the full structural index, finds the answer. A reader who wants to take up the theory of information dynamics finds the whole chain, with the exact status of each link.

**Status of each statement.** The document separates three categories and flags them throughout: *theorems* (exact under their assumptions), *postulates* (substantive, testable economic assumptions), and *empirical findings* (with the null they clear or fail). Table 1 gives the map.

| Item | Status | Empirical verdict | Section |
|---|---|---|---|
| Gaussian maximum entropy | theorem | — | 1 |
| Volume bridge (MDH, subordinated Brownian motion) | argument under assumptions | not tested here | 1 |
| Three-channel decomposition | exact identity | — | 2 |
| Structural index $\mathcal{J}$ as a divergence | exact identity | range clears both nulls | 2 |
| Downward entropy jumps | theorem (on average) | not directly estimable | 3 |
| Relaxation (P1) | postulate | half-life not identified | 3 |
| One-sided constraint (P2) | postulate | not established (H5) | 3 |
| Tail–correlation coupling | consequence | not identified (H6) | 3 |
| Dependence channel under stress | finding | **clears both nulls** | 4, 6 |
| KL ball, 1 nat $= \sqrt{2}\,\sigma$ | theorem | — | 5 |
| Minimal lift (rank one) | theorem (new) | — | 5 |
| Radius calibrated on the flow | rule | too mild against history | 5 |
| Block decomposition | identity (new here) | subject of the main paper | 6 |

Table 1: Map of the statements in this document and their status.

**Provenance.** Sections 1 to 3 and Appendices A to E take up, with updated cross-references, the text of the reference French version (`paper/main_fr.tex`), which supersedes `paper/main.tex` for the dynamics section and the appendices. Sections 4 to 8 are written for this document. Every number quoted comes from the same code (`python -m src.run_empirical`, two-hundred-replication nulls, Phase A); none was recomputed for this supplement.

## Notation

Two warnings before the table. First, differential entropy $h$ **depends on units**: $h(aX) = h(X) + \log|a|$, so expressing returns in percent rather than decimals shifts $h$ by a constant. Divergences ($D_{KL}(\cdot\,\|\,\cdot)$, $I$, $\mathcal{D}$, $\mathcal{J}$), by contrast, are invariant under any invertible change of variable: that is the reason for the structural index. Second, $K$ denotes kurtosis and "KL" stands for Kullback and Leibler; the two are unrelated.

| Symbol | Meaning | Nature |
|---|---|---|
| $p_t$ | true conditional return distribution at $t$ | — |
| $q_t$ | Gaussian with the same mean and covariance as $p_t$ | — |
| $h(\cdot)$ | differential entropy | nats, *unit-dependent* |
| $D_{KL}(p \,\|\, q)$ | Kullback–Leibler divergence | nats, invariant |
| $\mathcal{D}_t$ | entropy deficit $= D_{KL}(p_t \,\|\, q_t) \geq 0$ | nats, invariant |
| $\mathcal{D}_1(z)$ | deficit of a standardised scalar (negentropy) | nats, invariant |
| $H_t^{vol}$ | scale channel $= \sum_i \log \sigma_{i,t}$ | nats, *unit-dependent* |
| $H_t^{dep}$ | dependence channel $= \frac{1}{2} \log\det R_t \leq 0$ | nats, invariant |
| $H_t^{cov}$ | $H_t^{vol} + H_t^{dep}$; the constant $\frac{N}{2}\log 2\pi e$ is omitted | nats |
| $\mathcal{J}_t$ | structural index $= \mathcal{D}_t - H_t^{dep} \geq 0$ | nats, invariant |
| $I(X;Y)$ | mutual information | nats, invariant |
| $I_t^{(h)}$ | information flow $= D_{KL}(p_t \,\|\, p_{t-h})$ | nats |
| $S$, $K$ | skewness, excess kurtosis | dimensionless |
| $\Sigma_t = D_t R_t D_t$ | covariance, standard deviations, correlation | — |
| $\lambda_k$, $N_t^{\text{eff}}$ | eigenvalues of $R_t$, effective number of modes | $1 \leq N^{\text{eff}} \leq N$ |
| $\kappa$, $\sigma_{\mathcal{J}}$ | relaxation rate and diffusion coefficient of $\mathcal{J}$ | — |
| $\eta$ | radius of a stress scenario | nats |
| $N$ | number of assets | — |

# 1 From order flow to a Gaussian price: the volume bridge

## 1.1 The price as accumulated order flow

Over $[t, t+\Delta]$ let the market execute $M$ trades, trade $k$ carrying signed volume $\epsilon_k v_k$ with $\epsilon_k \in \{-1, +1\}$ the sign (buyer- or seller-initiated) and $v_k > 0$ the size. Write $V = \sum_k v_k$ for total volume and $Q = \sum_k \epsilon_k v_k$ for net signed order flow.

**Assumption 1** (Linear impact)**.** *The log-price responds linearly to net signed order flow,* $\Delta X_{t,\Delta} = X_{t+\Delta} - X_t = \lambda Q$*, with* $\lambda > 0$ *constant over the interval.*

This is Kyle's (1985) equilibrium: the market maker, unable to distinguish informed from uninformed flow, sets a linear pricing rule and $\lambda$ is the inverse depth. It is the definition of "price as a function of traded volume" that the rest of the section needs, and it is the weakest link in the chain; Section 1.5 says what happens when it fails.

**Assumption 2** (Exchangeable signs and a volume-proportional variance)**.** *Conditional on* $V$ *and on the information available at* $t$*, the market's belief about* $Q$ *is constrained only by its first two moments, with* $\mathbb{E}[Q \mid V] = m$ *and* $\mathrm{Var}(Q \mid V) = \varsigma^2 V$*.*

The variance form is what one gets when trade signs are exchangeable with bounded correlation and sizes have a stable mean: for independent signs, $\mathrm{Var}(Q) = \sum_k v_k^2 \approx \bar{v} V$. The economic content is that *volume, not time, is the natural measure of how much can happen.*

## 1.2 Maximum entropy in volume time

**Proposition 1** (Maximum entropy under first and second moments)**.** *Among absolutely continuous distributions on* $\mathbb{R}^N$ *with mean* $\mu$ *and covariance* $\Sigma \succ 0$*, the Gaussian* $\mathcal{N}(\mu, \Sigma)$ *uniquely maximises differential entropy, and* $h^{\max}(\Sigma) = \frac{1}{2}\log\big((2\pi e)^N \det\Sigma\big)$*.*

*Proof.* Let $q = \mathcal{N}(\mu, \Sigma)$ and let $p$ have the same first two moments and finite differential entropy. Since $\log q$ is an exact quadratic, its expectation depends only on those moments, so $\mathbb{E}_p[\log q] =$

$\mathbb{E}_q[\log q] = -h(q)$. Non-negativity of relative entropy gives $0 \le D_{KL}(p \,\|\, q) = -h(p) + h(q)$, hence $h(p) \le h(q)$, with equality iff $p = q$ almost everywhere. The argument requires no differentiability of $p$ and no prior existence of a maximiser. □

The variational route – a Lagrangian in the normalisation, mean and covariance constraints, whose first-order condition makes $\log p$ quadratic – explains *why* the maximiser is Gaussian, but the divergence argument is what establishes optimality and uniqueness.

Applying Proposition 1 to Assumption 2 in one dimension and mapping through Assumption 1:

$$\Delta X_{t,\Delta} \;\big|\; V \;\sim\; \mathcal{N}\big(\lambda m,\; \lambda^2 \varsigma^2 V\big)\,. \tag{1}$$

Equation (1) is the bridge. Three things about it deserve emphasis.

First, *the conditioning variable is volume, not time.* Maximum entropy does not deliver a variance; it delivers the least-committal distribution given one. Assumption 2 supplies the variance from an observable, and that observable is traded volume. Equation (1) is the mixture-of-distributions hypothesis of Clark (1973) and Tauchen & Pitts (1983), obtained here as a maximum-entropy statement rather than posited; Ané & Geman (2000) show empirically that returns measured in trade time are close to Gaussian even though calendar-time returns are not, which is the prediction (1) makes.

Second, *Gaussianity is conditional.* Unconditionally, $\Delta X$ is a scale mixture of normals over the random volume $V$, hence leptokurtic even when (1) holds exactly. This is the Mandelbrot & Taylor (1967) subordination mechanism, and it means fat tails are not by themselves evidence against maximum entropy.

Third, the multivariate extension is immediate: with $N$ assets, $\Delta\mathbf{X} \mid V \sim \mathcal{N}(\boldsymbol{\mu}V, \Sigma V)$, and Proposition 1 applies as stated.

### 1.3 From Gaussian increments to Brownian motion and to GBM

Let $\theta_t = \int_0^t v_s\, ds$ be the cumulative volume clock. Equation (1) gives the one-increment distribution; a process needs more.

**Assumption 3** (Temporal consistency)**.** *In the volume clock, increments over disjoint intervals are independent, sample paths are continuous, and the covariance of an increment is proportional to elapsed volume.*

Under Assumption 3 with constant coefficients there is a matrix $B$ with $BB^\top = \Sigma$ such that $X_\theta = X_0 + \mu\theta + BW_\theta$ in volume time, so in calendar time

$$X_t = X_0 + \mu\,\theta_t + B\,W_{\theta_t}, \tag{2}$$

a Brownian motion *subordinated* to the volume clock. If trading intensity is constant, $\theta_t = \bar{v}\,t$ and (2) is ordinary Brownian motion in calendar time; exponentiating and applying Itô,

$$\frac{dS_{i,t}}{S_{i,t}} = \big(a_{i,t} + \tfrac{1}{2}\sigma_{i,t}^2\big)\,dt + \sigma_{i,t}\,dW_{i,t}, \tag{3}$$

so the drift of the price is not the drift of the log-price – the same second-order correction that will reappear in Section 2 when we differentiate $\log\det R_t$.

It is worth being precise about what Assumption 3 does. By Monroe's (1978) theorem, *every* continuous local martingale is a time-changed Brownian motion, so subordination alone has no empirical content. The content is in the clock: (1) says the right clock is volume, and Assumption 3 says the market has no memory in that clock. Maximum entropy is what selects Brownian motion from the class Monroe's theorem allows.

### 1.4 The contrapositive

Proposition 1 runs both ways. Write $q_t$ for the Gaussian with the conditional mean and covariance of $p_t$. Then

$$\mathcal{D}_t \;\equiv\; h(q_t) - h(p_t) \;=\; D_{KL}(p_t \,\|\, q_t) \;\geq\; 0, \tag{4}$$

with equality if and only if $p_t$ is Gaussian. So: *prices are non-Gaussian exactly to the extent that their conditional distribution is not entropy-maximal*, and $\mathcal{D}_t$ measures the extent. In the language of Section 1, a positive $\mathcal{D}_t$ means constraints beyond the first two moments are binding – that somebody knows something the second moment does not encode.

Two sources produce a positive deficit, and separating them matters. A random volume clock produces a scale mixture, which is symmetric and leptokurtic. Binding directional information produces asymmetry. Section 2 splits $\mathcal{D}$ along exactly this line.

### 1.5 Where the bridge is weak

Assumption 1 is linear; the empirical impact of a metaorder is closer to square-root in size, which makes the mapping from flow to price concave and the induced return distribution thinner-tailed than the flow. Assumption 2 treats signs as exchangeable; order flow is in fact strongly autocorrelated, which inflates $\mathrm{Var}(Q)$ above $\varsigma^2 V$ and makes the volume clock an imperfect variance proxy. Neither failure changes the *form* of the argument – maximum entropy given a variance budget is still Gaussian – but both mean the volume clock is an approximation to the information clock rather than the thing itself. We use volume-time reasoning to motivate the framework and calendar-time estimation throughout the empirical work, so no result below depends on (1) holding exactly.

## 2 Three channels of entropy

### 2.1 Scale and dependence

Write $\Sigma_t = D_t R_t D_t$ with $D_t = \mathrm{diag}(\sigma_{1,t}, \ldots, \sigma_{N,t})$. Then $\det \Sigma_t = \left(\prod_i \sigma_{i,t}^2\right) \det R_t$ and

$$h(q_t) \;=\; \underbrace{\tfrac{N}{2}\log(2\pi e)}_{\text{constant}} \;+\; \underbrace{\sum_i \log \sigma_{i,t}}_{H_t^{vol}} \;+\; \underbrace{\tfrac{1}{2}\log\det R_t}_{H_t^{dep}}. \tag{5}$$

By Hadamard's inequality $H_t^{dep} \leq 0$, with equality iff $R_t = I$: *correlation always destroys entropy*, and $-H_t^{dep}$ is precisely the Gaussian multi-information (total correlation) of the return vector. A useful reading of the same quantity is spectral: with $\lambda_k$ the eigenvalues of $R_t$ and $\tilde{\lambda}_k = \lambda_k / N$, the participation number $N_t^{\text{eff}} = \exp\left(-\sum_k \tilde{\lambda}_k \log \tilde{\lambda}_k\right)$ is the effective number of independent risk modes; it equals $N$ for $R = I$ and falls toward 1 as one collective mode absorbs the variance.

Equation (5) is the ceiling, not the entropy. It is what the covariance matrix alone implies, and Proposition 1 says the truth sits below it.

### 2.2 Shape: the entropy deficit and what it is made of

Subtracting (4) from (5) gives the paper's decomposition, three channels and a constant:

$$h(p_t) \;=\; \tfrac{N}{2}\log(2\pi e) \;+\; \underbrace{H_t^{vol}}_{\text{scale}} \;+\; \underbrace{H_t^{dep}}_{\text{dependence}} \;-\; \underbrace{\mathcal{D}_t}_{\text{shape}}. \tag{6}$$

The shape channel is skewness and fat tails, and the Edgeworth expansion says so explicitly: for a standardised variable with skewness $S$ and excess kurtosis $K$ near Gaussianity,

$$J \;\approx\; \frac{S^2}{12} + \frac{K^2}{48}. \tag{7}$$

Equation (7) is the answer to the question "where would an entropy jump be visible?": in the third and fourth moments, additively, with an asymmetry channel and a tail channel. It is also the source of the sign problem that Section 3 has to solve, because $S$ enters squared – *negentropy cannot tell a crash from a melt-up.* As an estimator (7) is unusable on daily returns; Section 4 documents how badly and what we use instead.

### 2.3 The structural index

The scale channel is a nuisance for our purposes: it dominates the level of (6) numerically, it moves with volatility, and a change in units changes it. The other two channels are scale-free. Combining them, and using the exact chain rule $D_{KL}(p \,\|\, \prod_i q_i) = \sum_i D_{KL}(p_i \,\|\, q_i) + I(x)$ with $I$ the multi-information:

**Definition 1** (Structural information index)**.**

$$\mathcal{J}_t \;\equiv\; \mathcal{D}_t \;-\; H_t^{dep} \;=\; D_{KL}(p_t \,\|\, \textstyle\prod_i q_{i,t}) \;\geq\; 0, \tag{8}$$

the divergence of the conditional joint return distribution from a product of Gaussian marginals with the same scales.

$\mathcal{J}$ is zero if and only if returns are independent and Gaussian; it rises when assets couple, when marginals fatten, or when they skew. It is measured in nats, it is invariant to the level of volatility, and it is additive across its two channels, so one can always ask how much of a move came from dependence and how much from shape. It is the state variable of Section 3.

## 3 Information dynamics: one theorem, two postulates

### 3.1 Downward jumps are a theorem

Two distinct statements underpin the downward half of the postulated dynamics. Conflating them is an easy mistake, and earlier versions of this work made it; we therefore separate them explicitly.

Recall first the definition of the *mutual information* between the return vector $X$ and a signal $Y$:

$$I(X;Y) \;=\; D_{KL}(p_{XY} \,\|\, p_X \otimes p_Y) \;=\; \iint p(x,y)\, \log \frac{p(x,y)}{p(x)p(y)}\, dx\, dy, \tag{9}$$

that is, the divergence of the joint law from the product of the marginals: how far the joint law is from independence. It is non-negative, symmetric, zero if and only if $X \perp Y$, and, unlike $h$, invariant under invertible reparametrisation. For a scalar Gaussian pair with correlation $\rho$, $I = -\frac{1}{2}\log(1-\rho^2)$.

**Proposition 2** (Fall in average conditional entropy)**.** *Let $X$ be the return vector and $Y$ a signal. Then*

$$\mathbb{E}_Y[h(X \mid Y)] \;=\; h(X) - I(X;Y) \;\leq\; h(X),$$

*with equality if and only if $X \perp Y$.*

**Proposition 3** (Monotonicity of the ceiling under constraint)**.** *Let $\mathcal{C} \subseteq \mathcal{C}'$ be two constraint sets and $\mathcal{Q}(\cdot)$ the set of laws that satisfy them. Then $\mathcal{Q}(\mathcal{C}') \subseteq \mathcal{Q}(\mathcal{C})$ and hence*

$$\sup_{p \in \mathcal{Q}(\mathcal{C}')} h(p) \;\leq\; \sup_{p \in \mathcal{Q}(\mathcal{C})} h(p).$$

Both results are classical (Cover & Thomas, 2006) and they are not equivalent; the detailed proof and a worked Gaussian example are in Appendix E. Their scopes differ along two axes.

Proposition 2 *quantifies* the fall (exactly $I(X;Y)$ nats) but only *on average* over realisations of the signal. For a particular $y$, $h(X \mid Y = y)$ can exceed $h(X)$: a sufficiently confusing signal leaves the market more uncertain than before. One should therefore expect a jump component that is mainly, not exclusively, one-sided, which is what Section 4 reports.

Proposition 3, by contrast, holds *for every realisation*, but it concerns the maximum-entropy *ceiling*, not the actual entropy, and it quantifies nothing: it says that the ceiling cannot rise, not by how much it falls.

What the two say together, and what motivates everything that follows: the fall is *discontinuous*, because the signal arrives at a point in time, and its expected size in nats is the information content of the news, an amplitude measured on a scale that depends on neither the asset, the currency nor the units.

**An honesty caveat.** Neither proposition is used to compute anything in the empirical part. We measure $\mathcal{J}$, not $h(X) - \mathbb{E}_Y[h(X \mid Y)]$: estimating a differential entropy in dimension 20 on a 252-day window is out of reach (Appendix A). The link between the theorem and the statistics reported later is a line of reasoning, not an estimated identity, and the reader should treat it as such.

### 3.2 Two postulates

**Postulate 1** (Relaxation)**.** In the absence of new information, the structural index decays towards a long-run level at a constant rate: $d\mathcal{J}_t = -\kappa\left(\mathcal{J}_t - \bar{\mathcal{J}}\right) dt + \sigma_{\mathcal{J}}\, dW_t$ with $\kappa > 0$.

The economics behind P1 is that a constraint goes stale. The news is priced in, disagreement reforms, positions disperse, and the cross-section recovers the freedom the shock took away. Nothing in information theory delivers this; it is the substantive assumption of the paper, and $\kappa$ is the object to be estimated.

**Why an Ornstein–Uhlenbeck process and not something else?** The choice is not one of convenience: it is *forced* once three minimal requirements are imposed, and it is the only process that meets all three.

1. *Markov.* The future of $\mathcal{J}$ depends on the past only through its current level. This is the parsimony requirement: postulate no memory beyond what the current state already holds.

2. *Stationary.* $\mathcal{J}$ has a finite long-run level and variance. Without this the index drifts, it is not a state variable, and the regime comparisons of Section 4 are meaningless.

3. *Gaussian noise.* Between two information arrivals we claim to know nothing about the noise beyond its variance: this is the maximum-entropy principle applied to the noise itself, exactly as it was applied to returns in Section 1.

Doob (1942)'s theorem states that **the Ornstein–Uhlenbeck process is the only continuous-time process that is at once Markov, Gaussian and stationary**. The three requirements therefore leave no latitude on the diffusive part: it is determined. The jump term is added precisely to *break* Gaussianity where the framework says it fails, namely at information arrivals. The structure of (11) is thus an imposed skeleton plus a deliberately non-Gaussian component, which is exactly the class introduced by Barndorff-Nielsen & Shephard (2001).

**What remains arbitrary.** Three things, which must be stated rather than presented as if they followed from the theorem. The rate $\kappa$ is assumed *constant*: nothing rules out a crisis being absorbed faster, or slower, than ordinary news, and the data used here cannot tell. The pull-back is *linear* in the deviation from the long-run level, whereas a pull-back that accelerates far from the mean would be just as plausible. Finally, $\mathcal{J}$ is non-negative by construction, whereas an Ornstein–Uhlenbeck process with a Brownian term can turn negative: the Barndorff-Nielsen & Shephard (2001) specification avoids this by dropping the Brownian term and driving the process with a subordinator with positive jumps, which guarantees a stationary law supported on $\mathbb{R}^+$. Empirically the issue does not arise ($\mathcal{J}$ stays between 2.17 and 21.13 nats on the main panel and between 0.42 and 1.69 on the replication panel, far from the boundary), but formally one should either drop $\sigma_{\mathcal{J}}$ or write the process on $\log \mathcal{J}$.

Nor is the postulate innocent as a matter of measurement: an estimator with memory will manufacture decay even with no economic relaxation at all, which is why Section 4 measures the decay that a memoryless world already produces.

**Postulate 2** (Asymmetry of the constraint)**.** The constraint imposed by information is one-sided in the direction of losses, so that an entropy jump comes with negative skewness of the market factor.

**Why this postulate is logically necessary.** The index $\mathcal{J}$ is an *even* function of skewness: flipping the sign of skewness leaves its value unchanged. The reason is immediate. The shape channel is $\mathcal{D}_1(z) = D_{KL}(p_z \,\|\, \varphi)$, and the normal law $\varphi$ is symmetric; so replacing $z$ by $-z$ leaves the reference law unchanged, and the divergence too:

$$\mathcal{D}_1(-z) = \mathcal{D}_1(z), \qquad \text{and therefore} \qquad \mathcal{J}(-x) = \mathcal{J}(x). \tag{10}$$

The Edgeworth expansion (7) makes it visible at a glance: skewness $S$ enters squared.

The consequence is structural, not anecdotal. **A distribution with a long left tail (a crash) and its mirror image with a long right tail (a melt-up) have exactly the same $\mathcal{J}$.** The index measures the *size* of the information, never its direction. No refinement of the estimator will change this: it is a property of the object measured. If the paper is to say anything about direction, it must therefore add an external ingredient. P2 is that ingredient, and it is the smallest possible: a single statement, about the sign, added to a framework that by construction produces none.

**What it rests on.** Not on a mechanism we would assert, but on an established empirical regularity: the *leverage effect* in the sense of Black (1976), namely that negative returns are followed by larger rises in volatility than positive returns of the same size. It is one of the most robust stylised facts in empirical finance (Christie, 1982; Engle & Ng, 1993; Nelson, 1991). Translated into the language of this paper: for an equal quantity of information, the market's response is larger on the way down than on the way up, so a jump in $\mathcal{J}$ comes with negative market skewness.

We deliberately abstain from choosing among the competing causal explanations of this effect (financial leverage proper, volatility feedback, or asymmetric withdrawal of liquidity on the sell

side). The postulate depends on none of them: it asserts only the asymmetry, not its cause. It is separately testable (H5) and separately refutable, and Section 4 concludes that it is not established.

### 3.3 The resulting dynamics

Proposition 2 together with P1 gives a non-Gaussian Ornstein–Uhlenbeck process with one-sided jumps, of the class introduced by Barndorff-Nielsen & Shephard (2001) for stochastic volatility:

$$d\mathcal{J}_t \;=\; -\kappa\left(\mathcal{J}_t - \bar{\mathcal{J}}\right) dt \;+\; \sigma_{\mathcal{J}}\, dW_t \;+\; \int_{z>0} z\, \mathcal{P}(dt, dz). \tag{11}$$

The integral term needs careful reading, because its notation is compact.

**What $\mathcal{P}$ is not.** It is neither a Gaussian law nor a function. The letter $\mathcal{N}$ is reserved in this document for the normal law, so the jump measure carries a distinct symbol.

**What $\mathcal{P}$ is.** A *random counting measure*, known as a Poisson random measure. One can picture it as a random scatter of points in the (date, size) plane: each point is an information event, its abscissa is the date it occurs and its ordinate the size of the jump it causes. Formally, $\mathcal{P}(dt, dz)$ equals 1 if there is a jump in the time interval $[t, t+dt)$ whose size falls in $[z, z+dz)$, and 0 otherwise. The integral $\int_{z>0} z\,\mathcal{P}(dt, dz)$ then reads simply: "if there is a jump at date $t$, add its size $z$ to $\mathcal{J}$; otherwise add nothing". It is a compact way of writing $\sum_j z_j$, the sum of the jumps that occurred.

**What characterises it.** Two ingredients, combined in its *intensity measure* $\nu(dz)\, dt$, the expected number of jumps of size between $z$ and $z + dz$ during $dt$: an *arrival rate* $\lambda = \nu(\mathbb{R}^+)$, in jumps per unit time, and a *size distribution* $\nu(dz)/\lambda$. On the main panel, v1 estimates $\lambda \approx 5$ jumps per year, a mean size of 0.62 nat and a maximum of 3.21 nats.

**What $z$ is.** The size of an individual jump, in nats. Conceptually, it is the mutual information of the news that caused it, in the sense of Proposition 2. Operationally, it is the change in $\mathcal{J}$ picked up by the four-sigma filter of Section 4. The two coincide only approximately, and the paper does not claim to identify them.

**Why the integral runs over $z > 0$ only.** Because $\mathcal{J}$ measures an entropy *deficit* and, by Proposition 2, information can only increase that deficit in expectation. Within this framework nothing creates a *sudden increase* in entropy: entropy is restored only by the slow relaxation of the first term, never by a jump. Hence one-sided jumps.

Two caveats, however. Proposition 2 is a statement *on average*: a particularly confusing realisation can raise entropy, so a faithful specification would allow a small probability of negative jumps rather than forbid them. And the Poisson measure assumes independent arrivals, whereas the event study of v1 shows clear clustering of jumps; a self-exciting Hawkes-type process would be more faithful.

Note finally that the diffusion coefficient is written $\sigma_{\mathcal{J}}$ rather than $\eta$, the latter being reserved for the radius of stress scenarios (Section 5). Read on entropy itself rather than on $\mathcal{J}$, the sign flips: entropy falls by jumps and recovers by diffusion. The observable predictions are sharp, and they are what Section 4 tests: the distribution of $\Delta\mathcal{J}$ is right-skewed; large moves are up and small moves down; and after a positive jump the index decays with half-life $\ln 2/\kappa$.

### 3.4 Why fat tails, correlation and jumps arrive together

The framework yields a prediction that no volatility or correlation model produces on its own. It follows in four steps, without technical vocabulary.

1. Important news concerns *all* assets at once: the same information constrains each of them.
2. If the news is bad, all assets therefore fall *on the same day*.
3. That day also appears, taken in isolation, in each asset's own history: for that asset it is an unusually large loss.
4. Accumulating such days distorts each asset's return distribution: the loss side lengthens and thickens more than the gain side.

Step 2 is what the dependence channel measures: assets move together, measured correlation rises, $-H^{dep}$ rises. Step 4 is what the shape channel measures: extreme losses more numerous and larger than extreme gains, that is higher kurtosis and negative skewness, so $\mathcal{D}$ rises. The two channels rise together because they describe the same day seen from two places: the cross-section for one, each asset's distribution for the other.

In calm markets, notable moves are asset-specific: they do not align the cross-section and so have no reason to lift both channels at the same time.

The testable form is a *regime-dependent coupling*: the correlation between $-\Delta H^{dep}$ and $\Delta\mathcal{D}$ should be close to zero in calm markets and strongly positive under stress. This is H6, and it is the paper's answer to the question "does the link between fat tails, correlation and entropy jumps require an additional postulate?": no. Given Proposition 2, the coupling follows from the common nature of the news. Only the *sign* requires P2.

## 4 The H1–H8 empirical programme and what the nulls leave of it

This section summarises, without reproducing them, the estimators and tests of arXiv v1. It serves one purpose: to explain why, of all the theory in Sections 1 to 3, only the dependence channel makes it into the main paper.

### 4.1 The eight hypotheses

**H1 (regime signature).** Under stress, $H^{dep}$ falls, $\mathcal{D}$ rises, and so $\mathcal{J}$ rises.

**H2 (partial compensation).** $\Delta H_t^{dep} = -\beta\,\Delta H_t^{vol} + u_t$ with $\beta > 0$, and $\beta_{\text{stress}} > \beta_{\text{calm}}$.

**H3 (jump asymmetry).** Changes in $\mathcal{J}$ are one-sided, beyond what the same estimator produces on data with no temporal structure.

**H4 (relaxation).** $\mathcal{J}$ mean-reverts with a half-life longer than the estimator's own memory.

**H5 (skewness signature).** Positive jumps in $\mathcal{J}$ coincide with negative skewness of the market factor (Postulate P2).

**H6 (channel coupling).** Dependence and shape are weakly coupled in calm markets and strongly coupled under stress.

**H7 (predictive power).** Today's entropy improves, out of sample, the forecast of 21-day risk relative to volatility.

**H8 (composition of the crisis budget).** Grouped into episodes, jumps have the dependence channel as their dominant contributor more often than a memoryless resampling does.

**Estimation, in one sentence.** Rolling 252-day windows, Ledoit & Wolf (2004) shrinkage covariance, marginal deficit via the Hyvärinen (1998) approximation rather than the Edgeworth expansion (7) (unusable on daily returns), stress regime defined by a volatility threshold computed on the past only, jumps detected by a four-sigma filter on $\Delta\mathcal{J}$.

### 4.2 Why a calibrated null is indispensable

The theoretical point that governs the whole reading of the results is the following. $\mathcal{J}$ is a *convex, non-negative* functional of second- to fourth-order moments estimated over a window. When a fat-tailed observation enters the window, the estimated moments jump, and the convex functional jumps *up* whatever the sign of the observation. As the observation ages (exponential weighting) or leaves the window, the functional comes down gradually. The measuring device therefore manufactures, on its own, the sawtooth predicted by Proposition 2 and Postulate P1. An i.i.d. Gaussian simulation with no events at all already yields an apparent half-life of 35 days and 37 positive jumps against none negative under the four-sigma filter.

Hence two nulls, built by resampling the observed return matrix and running the identical pipeline two hundred times: the *i.i.d.* null (rows drawn independently; marginal fat tails and cross-sectional dependence kept, all chronology destroyed) and the 21-day *block* null (which also keeps volatility clustering). A statistic that clears both is evidence; one that clears only one is not identified.

### 4.3 The verdict

| | Statistic | Data | i.i.d. | block-21 | Verdict |
|---|---|---|---|---|---|
| H1 | regime gap $H^{dep}$ | $-0.99$ | $-0.21$ | $-0.51$ | **clears both** |
| H1 | regime gap $\mathcal{J}$ | 1.93 | 2.14 | 1.97 | fails both |
| H1 | regime gap $\mathcal{D}$ | 0.27 | 1.25 | 0.65 | wrong-signed |
| H2 | $\beta$ (stress) | 0.20 | 0.05 | 0.11 | fails both |
| H3 | positive jumps | 158 | 148.5 | 133.2 | block only |
| H3 | skewness of $\Delta\mathcal{J}$ | 12.26 | 14.80 | 12.97 | wrong-signed |
| H4 | half-life (days) | 120.8 | 92.1 | 162.8 | between the two |
| H4 | range of $\mathcal{J}$ | 18.95 | 10.05 | 13.77 | **clears both** |
| H5 | corr$(\Delta\mathcal{J}, \Delta S_{mkt})$ | $-0.24$ | $-0.04$ | $-0.07$ | fails both |
| H6 | coupling gap | 0.59 | 0.77 | 0.67 | wrong-signed |
| H7 | out-of-sample gain | none on two panels | | | rejected |
| H8 | share of dependence-led episodes | 46% | 19–33% | n.i. | i.i.d. only |

Table 2: Verdicts against the nulls (means over two hundred replications), main panel of twenty S&P 500 stocks, 1990–2022. Source: `paper/tables/tab_placebo_sp500.tex` and the H7–H8 sections of v1. n.i.: not identified, at every block length tested between 5 and 126 days.

Two statistics clear both nulls on the main panel, and only one replicates on Kenneth French's 48 industry portfolios: the regime gap of the dependence channel ($-3.65$ nats against $-0.56$ and

−2.46). Phase A then hardened it: it holds in devolatilised (Forbes–Rigobon) and rank-based form, the latter being the strongest against the block null ($p = 0.03$ on the S&P 500, 0.01 on the industries), whereas the Kritzman et al. (2011) absorption ratio fails the block null on both panels.

### 4.4 What the verdict says about the theory

One must separate what the verdict touches from what it does not.

**What is untouched.** The statements of Sections 1 and 2 are identities and theorems. So is Proposition 2. No empirical result can refute them; it can only show that the available estimators do not let us see them at work.

**What is touched.** The two postulates. P1 is supported only by amplitude (the range of $\mathcal{J}$), not by a relaxation rate: the half-life lies between the two nulls and its level depends on the null chosen. P2 is not established (H5). Channel coupling (H6), which Section 3 derived from the common nature of news, is reproduced and even exceeded by resampled data.

**Why the dependence channel survives.** A plausible explanation, consistent with Phase A, lies in the form of the estimator. $\frac{1}{2}\log\det R = \frac{1}{2}\sum_k \log\lambda_k$ weighs the whole spectrum of the correlation matrix, including the small eigenvalues, which collapse when the cross-section couples onto a few collective modes. A single large isolated return barely moves those small eigenvalues. The shape channel $\mathcal{D}$, by contrast, reacts strongly to a single extreme observation, which is exactly the signature the null reproduces. Concentration measures (absorption ratio, first-eigenvalue share) look only at the top of the spectrum, where the block null catches up with them.

**Consequence for the main paper.** The indicator keeps the one identified channel and drops $\mathcal{D}$, and hence $\mathcal{J}$. The theory of Sections 2 and 3 remains the justification of *why* this channel measures shared information (the Gaussian multi-information, equation (5) and Appendix B), but the postulated dynamics is not used.

## 5 The geometry of Kullback–Leibler-ball scenarios

The main paper keeps the simplest use of this geometry: pricing a published scenario in nats, which for a pure mean shift is half the squared Mahalanobis distance. This section gives the full construction of which that use is a special case, with the proofs that v1 only stated, and a new result on lifting a portfolio scenario to full dimension.

### 5.1 Divergence between two Gaussians

**Proposition 4** (Closed form). *For $p_1 = \mathcal{N}(\mu_1, \Sigma_1)$ and $p_0 = \mathcal{N}(\mu_0, \Sigma_0)$ on $\mathbb{R}^N$, with $\delta = \mu_1 - \mu_0$,*

$$D_{KL}(p_1 \,\|\, p_0) = \tfrac{1}{2}\Big[\underbrace{\delta^\top \Sigma_0^{-1}\delta}_{\textit{shift}} + \underbrace{\operatorname{tr}(\Sigma_0^{-1}\Sigma_1) - N - \log\det(\Sigma_0^{-1}\Sigma_1)}_{\textit{deformation}}\Big]. \tag{12}$$

*Proof.* $D_{KL}(p_1 \,\|\, p_0) = \mathbb{E}_{p_1}[\log p_1 - \log p_0]$. Both logarithms are quadratic; the trace trick of Appendix B gives $\mathbb{E}_{p_1}[(X-\mu_1)^\top \Sigma_1^{-1}(X-\mu_1)] = N$ and $\mathbb{E}_{p_1}[(X-\mu_0)^\top \Sigma_0^{-1}(X-\mu_0)] = \operatorname{tr}(\Sigma_0^{-1}\Sigma_1) + \delta^\top\Sigma_0^{-1}\delta$. Collect with the $\log\det$ terms. □

The shift term is half the squared Mahalanobis distance: this is the unit in which the main paper audits published scenarios. The deformation term is non-negative and zero if and only if $\Sigma_1 = \Sigma_0$; writing $\nu_k$ for the eigenvalues of $\Sigma_0^{-1}\Sigma_1$, it equals $\frac{1}{2}\sum_k(\nu_k - 1 - \log \nu_k)$, a sum of non-negative terms, one per eigendirection.

### 5.2 The projected ball and the exchange rate

**Definition 2** (Entropy-ball scenario)**.** Given today's conditional law $p_0$ and a severity $\eta > 0$, the severity-$\eta$ scenario for a risk functional $\rho$ is

$$p_\eta^\star = \arg \max_{p\,:\,D_{KL}(p\,\|\,p_0)\le\eta} \rho(p). \tag{13}$$

For a linear portfolio $w^\top r$ with mean $m_0$ and standard deviation $s_0$ under $p_0$, set $u = (m - m_0)/s_0$ and $v = s/s_0$. By Proposition 4 in one dimension,

$$g(u, v) = \tfrac{1}{2}\left(u^2 + v^2 - 1 - 2\log v\right). \tag{14}$$

**Proposition 5** (Solution of the projected ball)**.** *For $\rho = -m + c\,s$ with $c \ge 0$ (Gaussian VaR and expected shortfall are of this form), the maximum of $\rho$ subject to $g(u, v) \le \eta$ is attained on the boundary $g = \eta$ at a point satisfying*

$$u = -t, \qquad v - v^{-1} = c\,t, \qquad t > 0.$$

*For $c = 0$: $v = 1$ and $u = -\sqrt{2\eta}$.* A purely directional scenario of $\eta$ nats is a move of $\sqrt{2\eta}$ sigmas*; one nat buys $\sqrt{2} \approx 1.41$ sigma, and $k$ sigmas cost $k^2/2$ nats.*

*Proof.* $g$ is convex on $\mathbb{R} \times (0, \infty)$ (the sum of $u^2/2$ and $(v^2 - 1 - 2\log v)/2$, whose second derivative is $1 + v^{-2} > 0$) and the objective $s_0(-u + cv)$ is linear, so the Karush–Kuhn–Tucker conditions are sufficient. The objective has no interior maximum, so the constraint binds. The Lagrangian gives $-1 = \lambda u$ and $c = \lambda(v - v^{-1})$ with $\lambda > 0$; setting $t = 1/\lambda$ yields the stated system, and $t$ is pinned down by $g(-t, v(t)) = \eta$. For $c = 0$, $v - v^{-1} = 0$ forces $v = 1$ and $g = u^2/2 = \eta$. □

### 5.3 From the portfolio to full dimension

The ball (13) is solved on the portfolio's law, but a stress test must say how each asset moves. Two facts settle the question.

**Proposition 6** (Data processing)**.** *For any law $p$ on $\mathbb{R}^N$, with $p^w$ and $p_0^w$ the laws of $w^\top r$ under $p$ and $p_0$, $D_{KL}(p^w \,\|\, p_0^w) \le D_{KL}(p \,\|\, p_0)$.*

This is the data-processing inequality (Cover & Thomas, 2006) applied to the map $r \mapsto w^\top r$. The *ambient* price of a scenario (in full dimension) is therefore always at least its *portfolio* price (in projection). The classical-bundle table of v1 bears this out line by line (9.97 against 7.24 nats for the standard bundle).

**Proposition 7** (Minimal lift)**.** *Let $p_0 = \mathcal{N}(\mu_0, \Sigma_0)$, $m_0 = w^\top\mu_0$, $s_0^2 = w^\top\Sigma_0 w$, and a target $(m, s)$ with $s > 0$. Among Gaussians $\mathcal{N}(\mu_1, \Sigma_1)$ with $w^\top\mu_1 = m$ and $w^\top\Sigma_1 w = s^2$, the divergence $D_{KL}(p_1 \,\|\, p_0)$ is minimised by*

$$\mu_1 = \mu_0 + u\,s_0\,\beta, \qquad \Sigma_1 = \Sigma_0 + (v^2 - 1)\,s_0^2\,\beta\beta^\top, \qquad \beta = \frac{\Sigma_0 w}{s_0^2}, \tag{15}$$

*and the minimum equals exactly $g(u, v)$, the portfolio price.*

*Proof.* By Proposition 6, every candidate costs at least the price of its projection, which is $g(u, v)$ since the projection of a Gaussian is Gaussian with mean $m$ and standard deviation $s$. It therefore suffices to check that (15) attains this bound. We have $w^\top \beta = 1$, so $w^\top \mu_1 = m_0 + u s_0 = m$ and $w^\top \Sigma_1 w = s_0^2 + (v^2 - 1)s_0^2 = s^2$. In (12): $\delta^\top \Sigma_0^{-1} \delta = u^2 s_0^2 \, \beta^\top \Sigma_0^{-1} \beta = u^2$; $\operatorname{tr}(\Sigma_0^{-1}\Sigma_1) = N + (v^2 - 1)s_0^2 \, \beta^\top \Sigma_0^{-1} \beta = N + v^2 - 1$; and by the matrix determinant lemma, $\det(\Sigma_0^{-1}\Sigma_1) = 1 + (v^2 - 1)s_0^2 \, \beta^\top \Sigma_0^{-1} \beta = v^2$. Hence $D_{KL}(p_1 \,\|\, p_0) = \frac{1}{2}(u^2 + v^2 - 1 - 2\log v) = g(u, v)$. The matrix $\Sigma_1$ is positive definite for every $v > 0$. □

**Asset-by-asset reading.** The vector $\beta$ collects the assets' betas on the portfolio, $\beta_i = \operatorname{Cov}(r_i, w^\top r)/s_0^2$. Write $r = \beta f + e$ with $f = w^\top r$ and $e$ uncorrelated with $f$ under $p_0$. The scenario (15) multiplies the standard deviation of the portfolio factor $f$ by $v$, shifts its mean by $u$ sigmas, and leaves the law of $e$ unchanged. Each asset's variance rises by $(v^2 - 1)s_0^2\beta_i^2$ and its mean moves by $u s_0 \beta_i$: high-beta assets take the shock, and the correlations among them rise. It is a one-factor scenario, and it is the least informative of all scenarios that reach the target severity.

*Remark* (Correction to v1). arXiv v1 identified this minimum through its first-order conditions, $\Sigma_1^{-1} = \Sigma_0^{-1} - \lambda w w^\top$, and judged it "not interpretable asset by asset". By the Sherman–Morrison formula, this rank-one perturbation of the precision matrix *is* (15), a rank-one perturbation of the covariance along $\Sigma_0 w$, and it reads as above. Since v1 never uses it, no number depends on it.

**Corollary 1** (Cost of uniform dilation)**.** *Uniform dilation $\Sigma_1 = v^2 \Sigma_0$, the v1 default, has a deformation component equal to $N \cdot \frac{1}{2}(v^2 - 1 - 2\log v)$, that is $N$ times that of the minimal lift.*

*Proof.* Every eigenvalue of $\Sigma_0^{-1}\Sigma_1$ equals $v^2$; apply the spectral form of the deformation term. □

For the ten-year scenario on the main panel ($v = 1.63$, $N = 20$), the variance component is 0.34 nat at the minimum and $20 \times 0.34 \approx 6.8$ under uniform dilation, which is the figure in the v1 composition table. The corollary explains the v1 finding that the compositions calibrated on H1 (2.88 nats) and refined by H8 (0.71) cost *less* than the default: the default deforms every eigendirection equally, whereas a composition that concentrates the shock on the common mode moves towards the one-factor scenario (15), whose total cost is 0.41 nat including the mean shift. That uniform dilation costs $N$ times the minimum in variance is a general result; that the calibrated compositions fall between the two remains a finding on these panels, but it becomes expected rather than surprising.

### 5.4 Composing a given severity

**Proposition 8** (One severity, many compositions)**.** *If $w^\top \Sigma_a w = w^\top \Sigma_b w = s^2$, then for every $\alpha \in [0, 1]$, $\Sigma_\alpha = (1 - \alpha)\Sigma_a + \alpha \Sigma_b$ satisfies $w^\top \Sigma_\alpha w = s^2$.*

*Proof.* $w^\top \Sigma w$ is linear in $\Sigma$. □

Portfolio severity therefore does not constrain the composition. The ambient price, for its part, is convex in $\Sigma_1$ (the $-\log\det$ term is convex and the rest is linear), so along a mixture it is bounded by the mixture of prices, which guarantees that an intermediate composition never costs more than the dearer of its two endpoints.

**A physical ceiling on correlation.** With marginal volatilities fixed, $w^\top \Sigma w$ is maximised over valid correlation matrices at $\rho_{ij} = 1$, where it equals $(\sum_i w_i \sigma_i)^2$. On the main panel this ceiling is only about 1.5 times today's volatility: beyond the five-year rung, no rise in correlation alone suffices, and some rise in volatilities is unavoidable.

### 5.5 Calibrating the radius, and the limit of that calibration

The radius is read off the information flow of the projection, $I_t^{(h)} = D_{KL}\big(p_t^w \,\|\, p_{t-h}^w\big)$, whose quantiles (on non-overlapping 21-day blocks) convert a return period into a radius. Reading the radius off the $N$-dimensional flow overstates it by an order of magnitude, by Proposition 6. Calibrated this way, the v1 ladder gives a ten-year volatility multiplier of 1.63, while the ten-year quantile of the realised one-month volatility rise is 3.2. The ladder is a coherent, reproducible rule; its return periods are not validated. This is one of the reasons the main paper keeps the audit (pricing a given scenario) and drops generation (building a scenario with a given return period).

### 5.6 Reverse stress testing

Asking for the cheapest scenario that produces a given loss $L$ inverts the ball. In the projected Gaussian setting, Proposition 5 reads backwards. In the non-parametric setting, the solution is the exponential tilting of the sample (Csiszár, 1975), whose effective sample size collapses as the radius grows: at 9 nats it is 1.0 on the main panel, meaning that the tilted law is nothing but the worst observed day. Any implementation should report this effective size and switch to the Gaussian ball as soon as it drops below a few dozen.

## 6 The bridge to the main paper: block decomposition

This section connects the theory above to the indicator of the main paper. It establishes the identity behind the "contagion map" and shows where the dependence channel sits within the structural index.

### 6.1 The identity

Partition the $N$ series into $B$ blocks (for instance the CISS segments: money market, bonds, equities, financial intermediaries, foreign exchange), and write $R_{bb}$ for the diagonal blocks of the correlation matrix $R$ and $R_D = \mathrm{diag}(R_{11}, \ldots, R_{BB})$.

**Proposition 9** (Within/between-block decomposition)**.**

$$\underbrace{-\tfrac{1}{2}\log\det R}_{\textit{total dependence}} \;=\; \sum_{b=1}^{B} \underbrace{\big(-\tfrac{1}{2}\log\det R_{bb}\big)}_{\textit{within block } b} \;+\; \underbrace{\Big(-\tfrac{1}{2}\log\det\big(R_D^{-1/2} R R_D^{-1/2}\big)\Big)}_{\textit{between blocks}}, \tag{16}$$

*and each of the $B+1$ terms is non-negative. The between-block term is the Gaussian multi-information among blocks, $D_{KL}(\mathcal{N}(0,R) \,\|\, \bigotimes_b \mathcal{N}(0,R_{bb}))$; it is zero if and only if the blocks are pairwise uncorrelated.*

*Proof.* $R = R_D^{1/2}\big(R_D^{-1/2} R R_D^{-1/2}\big) R_D^{1/2}$ and the determinant is multiplicative, so $\log\det R = \log\det R_D + \log\det(R_D^{-1/2} R R_D^{-1/2})$, with $\log\det R_D = \sum_b \log\det R_{bb}$. Each $R_{bb}$ is a correlation matrix, so $-\frac{1}{2}\log\det R_{bb} \geq 0$ by Hadamard's inequality (Appendix B). The matrix $\tilde{R} = R_D^{-1/2} R R_D^{-1/2}$ has identity diagonal blocks, hence a unit diagonal, and Hadamard applies again. Finally Proposition 4 with $\mu_1 = \mu_0$, $\Sigma_1 = R$, $\Sigma_0 = R_D$ gives $\frac{1}{2}[\mathrm{tr}(R_D^{-1}R) - N - \log\det(R_D^{-1}R)]$, and $\mathrm{tr}(R_D^{-1}R) = \sum_b \mathrm{tr}(I) = N$. □

**Two blocks: canonical correlations.** For $B = 2$, writing $C = R_{11}^{-1/2} R_{12} R_{22}^{-1/2}$ and $\rho_1, \ldots, \rho_k$ for its singular values (the canonical correlations between the two blocks), $\det \tilde{R} = \det(I - CC^\top) = \prod_j (1 - \rho_j^2)$ and the between-block term equals $-\frac{1}{2} \sum_j \log(1 - \rho_j^2)$. This is the Gaussian mutual information between the two segments, the sum of the information carried by each canonical pair. For $B > 2$, identity (16) can be applied in nested fashion and lets one attribute contagion to pairs or groups of segments by the chain rule.

### 6.2 Where the dependence channel sits in the structural index

The chain rule of Appendix D writes $\mathcal{J} = \sum_i \mathcal{D}_1(x_i) + I(x)$. Replacing the multi-information $I(x)$ by its Gaussian part $-H^{dep}$ and applying (16):

$$\mathcal{J} \;\approx\; \underbrace{\sum_i \mathcal{D}_1(x_i)}_{\text{shape}} \;+\; \underbrace{\sum_b \left(-\tfrac{1}{2} \log\det R_{bb}\right)}_{\text{within}} \;+\; \underbrace{\left(-\tfrac{1}{2} \log\det \tilde{R}\right)}_{\text{between}}. \tag{17}$$

The main paper keeps the last two terms and drops the first, which is the one the null reproduces (Section 4). The entropic CISS is thus the structural index stripped of its shape channel, and its decomposition is exact, with no approximation, because it concerns only the Gaussian part.

### 6.3 Three properties useful for the indicator

**Scale invariance.** The terms of (16) depend only on $R$: they are invariant to any series-by-series rescaling, which allows markets with very different volatilities to be aggregated without ad hoc weighting. The Forbes & Rigobon (2002) devolatilisation therefore acts not on the formula but on the bias of the *estimator* of $R$ under heteroskedasticity.

**Rank version.** Replacing $R$ by the correlation of normal scores $\Phi^{-1}(\hat{F}_i(x_i))$ estimates the multi-information of a Gaussian copula. The quantity becomes invariant to any monotone transformation of the marginals, which explains its robustness to tails observed in Phase A.

**Normalisation in $N$.** For a constant correlation $\rho$, $\det R = (1-\rho)^{N-1}\big(1 + (N-1)\rho\big)$, so $-H^{dep}$ grows roughly linearly in $N$. Dividing by $N - 1$, or reporting the equivalent constant correlation $\rho^\star$ that gives the same determinant, makes the indicator comparable across panels. For small $\rho^\star$, $-H^{dep} \approx \frac{1}{4} N(N-1)\rho^{\star 2}$.

## 7 Monitoring indicators and insurance solvency

**Two indicators from v1.** The jump/diffusion split delivered two daily quantities: the standardised positive-jump component of $\mathcal{J}$ (how much information arrived today) and $\mathcal{J}_t$ relative to its rolling maximum (how much has not yet been absorbed). Neither improves the volatility forecast (H7), and both rest on statistics the null does not identify. They are not carried over.

**Insurance.** Three links remain valid in principle. An entropy-ball scenario produces a complete stressed law, a natural object for the ORSA. The decomposition of classical bundles bears on the Solvency II standard formula, whose diversification credit rests on a fixed correlation matrix. And entropic risk measures are coherent (Ahmadi-Javid, 2012; Artzner et al., 1999). Nothing has been tested on liabilities, however, and the short, low-frequency panels typical of insurance data would strain every estimator used here.

# 8 Open questions

1. **Self-exciting arrivals.** The jump clustering seen in the v1 event study contradicts the Poisson measure in (11). A Hawkes process is the next specification.

2. **Positivity.** $\mathcal{J} \geq 0$ whereas an Ornstein–Uhlenbeck process with a Brownian term can turn negative. One should either drop $\sigma_{\mathcal{J}}$ (the pure form of Barndorff-Nielsen & Shephard 2001) or model $\log \mathcal{J}$.

3. **Non-linear tail dependence.** $\mathcal{J}$ and the dependence channel see only the Gaussian part of the multi-information. Estimating the full copula multi-information (for example by nearest neighbours on ranks) would measure what is missing, at the price of a much higher estimation variance.

4. **Tail of the information flow.** Beyond about twenty years, the quantiles of $I^{(h)}$ are extrapolated. A generalised Pareto fit would be the disciplined correction.

5. **Minimal lift and published scenarios.** Proposition 7 provides a yardstick: for any published scenario, the ratio of its ambient price to the minimal price $g(u, v)$ of its projection measures how much information it spends outside the direction that matters to the portfolio. This extends the audit of the main paper directly.

# A Units, invariance, and what can be computed

**The nat.** Entropy $H = -\sum_i p_i \log p_i$ is defined only once the base of the logarithm is chosen, and that choice *is* the choice of unit. With $\log_2$ one gets bits (shannons), with $\log_{10}$ dits, and with the natural logarithm *nats*. One nat is the information delivered by an observation that divides the space of possibilities by $e \approx 2.718$, where a bit divides it by two. The conversion is $1 \text{ nat} = 1/\log 2 \approx 1.4427$ bit. This document uses the natural logarithm throughout: entropies, deficits, divergences, scenario radii and the structural index are in nats.

Three operational readings, from the most useful to the most abstract. First the exchange rate established in Section 5: a purely directional scenario of $\eta$ nats is a move of $\sqrt{2\eta}$ sigmas, so $k$ sigmas cost $k^2/2$ nats. Next, coding: $D_{KL}(p \,\|\, q)$ nats is the expected excess code length when data with law $p$ are encoded with a code optimal for $q$. Finally, hypothesis testing: a divergence of one nat per observation means that the likelihood ratio in favour of $p$ grows on average by a factor $e$ with each observation.

**$h$ depends on units, divergences do not.** The differential entropy of a continuous variable satisfies, for any invertible affine transformation,

$$h(aX + b) \;=\; h(X) + \log|a|, \qquad \text{and in dimension } N, \quad h(AX + b) = h(X) + \log|\det A|. \tag{18}$$

Expressing returns in percent rather than decimals therefore adds $N \log 100$ to $h$: the *level* of $h$, and hence that of $H^{vol}$ and $H^{cov}$, has no interpretable absolute value. By contrast $D_{KL}(p \,\|\, q)$, $I(X;Y)$, $\mathcal{D}$ and $\mathcal{J}$ are invariant under any invertible change of variable, the Jacobians cancelling between numerator and denominator of the logarithm. *A figure in nats is comparable across datasets only if it is a divergence.* This is the structural reason the paper builds a scale-free index.

**How is $h(X)$ computed?** In the Gaussian case, in closed form (Appendix B). In general, it cannot be without knowing $p$. Non-parametric estimators exist (Vasicek (1976) spacings, Kozachenko–Leonenko nearest neighbours, kernel methods) but all degrade quickly beyond three or four dimensions. Estimating a differential entropy in dimension 20 on a 252-day window is out of reach.

The paper therefore *never* computes $h(p)$. It computes $h(q)$ in closed form from $\widehat{\Sigma}$, and the deficit $\mathcal{D}$ through an approximation of the *difference* of marginal entropies. A difference of entropies is dimensionless and estimable; a bare differential entropy is not. The whole architecture (decompose rather than estimate directly) follows from this constraint.

# B Gaussian entropy: proof of equation (5)

*Step A: entropy of a multivariate Gaussian.* For $q = \mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^N$,

$$\log q(x) = -\tfrac{1}{2}\left[N \log 2\pi + \log\det\Sigma + (x-\mu)'\Sigma^{-1}(x-\mu)\right],$$

so

$$h(q) = -\mathbb{E}_q[\log q] = \tfrac{1}{2}N\log 2\pi + \tfrac{1}{2}\log\det\Sigma + \tfrac{1}{2}\,\mathbb{E}\big[(X-\mu)'\Sigma^{-1}(X-\mu)\big].$$

The last expectation follows from the trace trick: a scalar equals its trace, and the trace is cyclic, hence

$$\mathbb{E}\big[(X-\mu)'\Sigma^{-1}(X-\mu)\big] = \mathbb{E}\big[\mathrm{tr}\big(\Sigma^{-1}(X-\mu)(X-\mu)'\big)\big] = \mathrm{tr}(\Sigma^{-1}\Sigma) = \mathrm{tr}(I_N) = N.$$

Collecting terms, $h(q) = \tfrac{1}{2}\left[N\log 2\pi + \log\det\Sigma + N\right] = \tfrac{1}{2}\log\big((2\pi e)^N \det\Sigma\big)$. □

*Step B: factorising the determinant.* With $\Sigma = DRD$, $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_N)$ and $R$ the correlation matrix, multiplicativity of the determinant gives $\det\Sigma = \det(D)\det(R)\det(D) = \left(\prod_i \sigma_i\right)^2 \det R$. □

*Step C: substitution.* $\tfrac{1}{2}\log\det\Sigma = \tfrac{1}{2}\log\Big[\left(\prod_i \sigma_i\right)^2 \det R\Big] = \sum_i \log\sigma_i + \tfrac{1}{2}\log\det R$, hence $h(q) = \tfrac{N}{2}\log(2\pi e) + \sum_i \log\sigma_i + \tfrac{1}{2}\log\det R$, which is equation (5). □

**Sign of the dependence channel.** Hadamard's inequality gives $\det R \le \prod_i R_{ii} = 1$ for a positive semi-definite matrix. A more telling proof goes through the spectrum: $\det R = \prod_k \lambda_k$ with $\sum_k \lambda_k = \mathrm{tr}\, R = N$, and the arithmetic–geometric mean inequality gives $\prod_k \lambda_k \le \left(\sum_k \lambda_k / N\right)^N = 1$, with equality if and only if every eigenvalue equals 1, that is $R = I$. Hence $H^{dep} \le 0$: *correlation always destroys entropy.*

In the bivariate case, $\det R = 1 - \rho^2$ and $H^{dep} = \tfrac{1}{2}\log(1-\rho^2)$; for $\rho = 0.9$ this is $-0.83$ nat. Note that $-H^{dep}$ is then exactly the mutual information between the two assets: correlation is shared information, hence destroyed entropy.

**Reading note on the tables.** The $H^{cov}$ columns of the v1 results tables report $H^{vol} + H^{dep}$ and **omit the constant** $\tfrac{N}{2}\log(2\pi e)$, which equals 28.38 nats for $N = 20$. This constant offset is harmless since only differences between regimes and changes over time are interpreted, but it explains why the reported levels are negative.

## C The entropy deficit: why $\mathcal{D} = h(q) - h(p)$

The Kullback–Leibler divergence between a law $p$ and a reference law $q$ is $D_{KL}(p \,\|\, q) = \int p \log(p/q)$, non-negative and zero if and only if $p = q$ almost everywhere. It is neither symmetric nor equipped with a triangle inequality: it is not a distance, hence the word divergence.

**Proposition 10.** *If $q = \mathcal{N}(\mu, \Sigma)$ has the first two moments of $p$, then $D_{KL}(p \,\|\, q) = h(q) - h(p)$.*

*Proof.* $D_{KL}(p \,\|\, q) = -h(p) - \mathbb{E}_p[\log q(X)]$. Now $\log q$ is exactly quadratic in $x$, so its expectation depends only on the mean and covariance of the law under which it is evaluated. Since $p$ and $q$ share these two moments, $\mathbb{E}_p\big[(X-\mu)'\Sigma^{-1}(X-\mu)\big] = \operatorname{tr}(\Sigma^{-1}\Sigma) = N = \mathbb{E}_q[\cdot]$, hence $\mathbb{E}_p[\log q] = \mathbb{E}_q[\log q] = -h(q)$ and so $D_{KL}(p \,\|\, q) = h(q) - h(p)$. □

The moment-matching assumption is essential: without it the identity fails. This is the whole content of Proposition 1, read as a measurement rather than an existence theorem. Economically, $\mathcal{D}$ is the price in nats of the lie "the market is Gaussian": the apparent uncertainty that the Gaussian description overstates, because the true law is more concentrated than its covariance alone suggests.

## D The structural index is a divergence

**Proposition 11.** *With $q_i = \mathcal{N}(\mu_i, \sigma_i^2)$ the Gaussian marginals with the same scales as $p$,*

$$D_{KL}(p \,\|\, \textstyle\prod_i q_i) \;=\; \mathcal{D} - H^{dep} \;=\; \mathcal{J}.$$

*Proof.* By definition,

$$\begin{aligned} D_{KL}(p \,\|\, \textstyle\prod_i q_i) &= -h(p) - \mathbb{E}_p[\textstyle\sum_i \log q_i(X_i)] = -h(p) + \displaystyle\sum_i \big[\tfrac{1}{2}\log(2\pi\sigma_i^2) + \tfrac{1}{2}\big] \\ &= -h(p) + \sum_i \log \sigma_i + \tfrac{N}{2}\log(2\pi e), \end{aligned}$$

using $\mathbb{E}_p[(X_i - \mu_i)^2] = \sigma_i^2$, true by construction. Substituting equation (6), namely $h(p) = \frac{N}{2}\log(2\pi e) + \sum_i \log \sigma_i + H^{dep} - \mathcal{D}$, the scale terms and the constant cancel and $\mathcal{D} - H^{dep}$ remains. □

This is an exact identity, not an approximation, and it is what justifies Definition 1. Removing the scale channel from (6) might look like a fudge; the object that remains turns out to be a proper Kullback–Leibler divergence, and so inherits three properties for free: it is non-negative; it is zero if and only if returns are independent *and* Gaussian, which provides a well-defined rather than arbitrary reference point; and it is invariant to asset-by-asset changes of units, since $p$ and $\prod_i q_i$ transform in the same way.

It is useful to set this identity beside the general chain rule $D_{KL}(p \,\|\, \prod_i q_i) = \sum_i D_{KL}(p_i \,\|\, q_i) + I(x)$, where $I(x) = D_{KL}(p \,\|\, \prod_i p_i)$ is the multi-information. The first term is the sum of marginal deficits $\sum_i \mathcal{D}_1(x_i)$, the second the total dependence. The estimator used in v1 estimates the first term with the Hyvärinen (1998) approximation and replaces the second with its linear part $-H^{dep}$; non-linear tail dependence is therefore omitted from $\mathcal{J}$ and measured separately through a co-exceedance rate.

## E Mutual information: the two propositions and an example

**Proof of Proposition 2.** By definition of conditional entropy, $\mathbb{E}_Y[h(X \mid Y)] = h(X,Y) - h(Y)$, and by (9), $I(X;Y) = h(X) + h(Y) - h(X,Y)$. Adding the two, $\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y)$. Non-negativity of $I$ follows from that of the Kullback–Leibler divergence, of which it is a special case, with equality if and only if $p_{XY} = p_X \otimes p_Y$.

**Proof of Proposition 3.** Immediate: an extra constraint can only shrink the feasible set, and the supremum of a functional over a smaller set cannot increase.

**Why the two are not equivalent.** The first quantifies but holds only on average; the second holds realisation by realisation but concerns the ceiling and quantifies nothing. A discrete counterexample suffices to show that a particular realisation can *raise* entropy. Let $X$ be binary with $\mathbb{P}(X = a) = 0.99$, so $H(X) = 0.056$ nat. Let a signal $Y$, with probability 0.98, remove all doubt ($H(X \mid Y = 0) = 0$), and with probability 0.02 lead to $\mathbb{P}(X = a \mid Y = 1) = 1/2$, so $H(X \mid Y = 1) = \log 2 = 0.693$ nat, twelve times the prior entropy. The average is nonetheless $0.98 \times 0 + 0.02 \times 0.693 = 0.014$ nat, well below 0.056, and $I(X;Y) = 0.042$ nat. This is exactly why the paper predicts a jump component that is *mainly*, not exclusively, one-sided.

**Worked Gaussian example.** Let $X \sim \mathcal{N}(0, \sigma^2)$ be tomorrow's return and $Y = X + \varepsilon$ a noisy signal, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ independent. Then

$$I(X;Y) = \tfrac{1}{2} \log\left(1 + \frac{\sigma^2}{\sigma_\varepsilon^2}\right), \qquad \mathrm{Var}(X \mid Y) = \frac{\sigma^2 \sigma_\varepsilon^2}{\sigma^2 + \sigma_\varepsilon^2}, \tag{19}$$

the posterior variance not depending on the realisation $y$. With $\sigma = 1\%$ a day:

| Signal quality | $\sigma_\varepsilon$ | $I(X;Y)$ (nats) | posterior vol. | $\sqrt{2\eta}$ equivalent |
|---|---|---|---|---|
| sharp | 0.5% | 0.805 | 0.45% | $1.27\sigma$ |
| moderate | 1.0% | 0.347 | 0.71% | $0.83\sigma$ |
| vague | 3.0% | 0.053 | 0.95% | $0.32\sigma$ |

A factor of fifteen separates the size of the entropy jump caused by decisive news from that caused by a rumour. The last column translates each jump onto the severity scale of Section 5: it is the directional move a scenario of the same radius would produce.

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